dot-jax: Differentiable Diffuse Optical Tomography with Automatic Differentiation, GPU Acceleration, and Standard-Space Head Modelling Applied to High-Density DOT Naturalistic Viewing

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Abstract

We present **dot-jax**, an open-source JAX/Equinox toolbox for continuous-wave diffuse optical tomography (CW-DOT) and functional near-infrared spectroscopy (fNIRS). dot-jax reimplements the redbirdpy finite element forward model as a fully differentiable, JIT-compiled, GPU-ready pipeline, enabling automatic Jacobian computation through the complete forward–inverse chain. The toolbox provides: (1) analytical CW fluence solutions cross-validated against redbirdpy, (2) chromophore extinction tables with Beer–Lambert spectral unmixing, (3) FEM system matrix assembly with Robin boundary conditions, (4) dense and sparse forward solvers via Lineax, (5) GCV-optimised Tikhonov image reconstruction, and (6) a comprehensive signal processing chain including motion artifact correction, channel quality control, short-separation regression, and spatial smoothing.

We validate dot-jax against redbirdpy at every computational layer (272 tests) and demonstrate the complete pipeline on the ds004569 high-density DOT dataset [Sherafati et al., 2025] from OpenNeuro: 96 sources, 92 detectors, 3,684 channels at 750/850 nm, with naturalistic movie viewing. Using a standard-space MNI152 atlas head mesh, we reconstruct oxyhemoglobin (HbO_2) concentration images at 1 Hz and compare to the published NeuroDOT tomographic reconstructions.

1 Introduction

Diffuse optical tomography (DOT) and functional near-infrared spectroscopy (fNIRS) exploit the relative transparency of biological tissue to near-infrared light (650–950 nm) to measure hemodynamic changes non-invasively [Jöbsis, 1977, Scholkmann et al., 2014]. In this spectral window, photons penetrate several centimetres into tissue, scattered by cellular membranes and mitochondria, absorbed primarily by oxy- and deoxyhemoglobin (HbO_2, Hb), water, and lipids. The wavelength-dependent absorption provides a spectroscopic contrast mechanism: multi-wavelength measurements enable separation of chromophore concentration changes via the modified Beer–Lambert law [Cope and Delpy, 1988].

High-density DOT (HD-DOT) arrays with overlapping measurement channels achieve spatial resolution comparable to fMRI [Eggebrecht et al., 2014], while retaining the temporal resolution, portability, and ecological validity advantages of optical methods. The combination of dense spatial sampling with tomographic image reconstruction transforms fNIRS from a surface mapping technique into a three-dimensional brain imaging modality.

The computational pipeline for DOT — from photon transport modelling through finite element assembly to regularised image reconstruction — has been implemented in several established packages: TOAST (Schweiger & Arridge), NIRFAST [Dehghani et al., 2009], and NeuroDOT [Eggebrecht et al., 2012]. These tools have enabled clinical and neuroscience applications but share a common limitation: they treat the forward model as a black box for gradient computation, relying on explicit Jacobian assembly via the adjoint method rather than automatic differentiation. This prevents end-to-end gradient flow through the complete pipeline and limits integration with modern machine learning frameworks.

1.1 Differentiable generative modelling with the JAX ecosystem

Generative models in neuroimaging. The generative modelling approach to neuroimaging inverts a forward (biophysical) model to infer latent parameters from observed data. Rather than treating brain images as static maps, generative models simulate the physical processes that produce the measurements — photon transport, hemodynamic coupling, optical signal formation — and fit the model parameters to match empirical recordings. In DOT, this approach is the foundation: the forward model simulates photon diffusion through tissue, and image reconstruction inverts this model to recover spatial maps of optical properties from boundary measurements [Arridge, 1999].

Simulation-based inference and neural differential equations. Two developments in machine learning transform what is computationally feasible for DOT. First, **simulation-based inference** (SBI) [Cranmer et al., 2020] replaces likelihood evaluation with learned neural density estimators trained on simulated data, enabling Bayesian posterior estimation for models with intractable likelihoods. For DOT, where the mapping from tissue optical properties to boundary measurements involves solving a PDE on a complex geometry, SBI offers a path to uncertainty quantification that traditional methods cannot provide. Second, **neural ordinary differential equations** [Chen et al., 2018] and physics-informed neural networks (PINNs) [Raissi et al., 2019] embed differentiable computation within physical solvers, enabling learned dynamics that respect the diffusion equation while remaining fully differentiable.

The JAX + Equinox + Lineax stack. dot-jax is built on JAX [Bradbury et al., 2018] with Equinox [Kidger and Garcia, 2021] for pytree-based modules and Lineax for linear solvers — the same ecosystem developed by Kidger [Kidger, 2022] for neural differential equations. JAX provides `jax.grad` (automatic differentiation through arbitrary Python), `jax.vmap` (automatic vectorisation), `jax.jit` (XLA compilation to GPU/TPU), and `jax.lax.scan` (efficient sequential computation). The result is that the entire DOT forward model — from optical property specification through FEM assembly and linear solve to detector measurement extraction — compiles into a single XLA computation graph through which gradients flow end-to-end:

$$\hat{\mathbf{y}} = \mathbf{R}_{\text{det}}^{\top}\big(\mathbf{K}(\mu_s') + \mathbf{M}(\mu_a) + \mathbf{C}(R_{\text{eff}})\big)^{-1}\mathbf{R}_{\text{src}} \tag{1}$$

where $\mathbf{K}$, $\mathbf{M}$, $\mathbf{C}$ are the stiffness, mass, and Robin boundary matrices, $\mathbf{R}_{\text{src/det}}$ are barycentric source/detector vectors, and the entire expression is differentiable with respect to μ_a and μ_s' via `jax.grad`. This enables gradient-based fitting, automatic Jacobian computation, and integration with simulation-based inference frameworks — capabilities absent from existing DOT software.

2 Dataset

2.1 OpenNeuro ds004569: high-density DOT naturalistic viewing

We use the ds004569 dataset [Sherafati et al., 2025] from OpenNeuro, comprising HD-DOT recordings from 58 healthy adults (aged 18–76) watching a 10-minute clip from "The Good, the Bad and the Ugly" (1966). The dataset provides:

- **Raw data** in SNIRF [fNIRS community, 2023] and NeuroDOT `.mat` formats, organised following BIDS-fNIRS [Luke et al., 2021]

- **96 sources, 92 detectors** in a high-density cap (BigCap1), sampled at 10 Hz
- **3,684 channels** at two wavelengths: 750 and 850 nm
- **Source–detector separations** at 1st–4th nearest neighbours (13, 30, 39, 47 mm)
- **Published NIfTI reconstructions** in Talairach space at 3 mm isotropic, 1 Hz, produced by NeuroDOT with the "Amat" reconstruction algorithm and 13 mm spatial smoothing

We focus on **sub-02**, the subject with the most data (7 sessions, 3.8 GB), using session 01 for pipeline demonstration.

Table 1: ds004569 sub-02/ses-01 acquisition parameters.

Parameter	Value
Sampling frequency	10.00 Hz
Channel count	3684
Source optodes	96
Detector optodes	92
Recording duration	553.6 s
Cap configuration	96X92
Mean SNR	19.5
Wavelengths	750, 850 nm

3 Methods

3.1 Photon diffusion forward model

Continuous-wave photon transport in tissue is modelled by the steady-state diffusion equation [Ishimaru, 1978, Patterson et al., 1989]:

$$-\nabla \cdot (D\,\nabla\Phi) + \mu_a\,\Phi = S \tag{2}$$

where $\Phi(\mathbf{r})$ is the photon fluence rate, $D = 1/[3(\mu_a + \mu_s')]$ is the diffusion coefficient, μ_a is the absorption coefficient, μ_s' is the reduced scattering coefficient, and S is the source term. The Robin (type III) boundary condition [Haskell et al., 1994] at the tissue–air interface is:

$$\Phi + 2A\,D\,\frac{\partial\Phi}{\partial\hat{n}} = 0 \qquad \text{where} \quad A = \frac{1 + R_{\text{eff}}}{1 - R_{\text{eff}}} \tag{3}$$

and R_{eff} is the effective Fresnel reflection coefficient computed via numerical integration over all angles of incidence.

3.2 Finite element discretisation

Following Arridge et al. [1993] and Schweiger et al. [1995], we discretise (2) with linear tetrahedral elements on an atlas head mesh. The FEM system is:

$$\underbrace{(\mathbf{K} + \mathbf{M} + \mathbf{C})}_{\mathbf{A}}\,\boldsymbol{\Phi} = \mathbf{b} \tag{4}$$

with stiffness $K_{ij} = \sum_e D_e \int_{\Omega_e} \nabla\phi_i \cdot \nabla\phi_j\,dV$, consistent mass $M_{ij} = \sum_e \mu_{a,e} \int_{\Omega_e} \phi_i\,\phi_j\,dV$, and Robin boundary $C_{ij} = \frac{1}{2A}\int_\Gamma \phi_i\,\phi_j\,dS$. The gradient products $\nabla\phi_i \cdot \nabla\phi_j$ are precomputed per element via the deldotdel operator, cross-validated against redbirdpy.

3.3 Atlas-based head mesh

We generate a standard-space head mesh from the MNI152 ICBM 2009c atlas via:

1. Tissue probability maps (CSF, GM, WM at 1 mm) fetched via nilearn
2. Brain mask extraction by thresholding
3. Marching cubes surface extraction (scikit-image)
4. Interior point sampling and Delaunay tetrahedralisation (scipy)
5. Element pruning (remove exterior/degenerate elements)
6. FEMMesh construction with precomputed operators

Optode positions (in Talairach/MNI coordinates from the SNIRF file) are projected to the mesh surface via nearest-neighbour lookup.

3.4 Signal processing pipeline

The raw CW intensity data undergo the following preprocessing:

Channel quality control. Channels are pruned based on three criteria: signal-to-noise ratio ($\mathrm{SNR} = \bar{I}/\sigma_I < 2$ rejected), coefficient of variation ($\mathrm{CV} = \sigma_I/\bar{I} > 0.5$ rejected), and saturation (fraction of samples at rail > 0.999 rejected).

Optical density conversion. Raw intensity is converted to optical density change: $\Delta\mathrm{OD}(t) = -\ln\big(I(t)/\bar{I}\big)$.

Motion artifact correction. Spike artifacts are detected where $|dI/dt| > 5\sigma$ and corrected via cubic spline interpolation through clean segments.

Short-separation regression. Channels with source–detector distance < 10 mm are used as regressors for superficial physiology [Saager and Berger, 2005, Brigadoi and Cooper, 2015]. A GLM with the mean short-channel signal as regressor is applied to each long channel.

Bandpass filtering and downsampling. A 5th-order Butterworth bandpass (0.01 Hz to 0.5 Hz) removes physiological noise and drift. Block averaging downsamples from 10 Hz to 1 Hz.

3.5 Image reconstruction

Sensitivity matrix. The absorption Jacobian is computed via the adjoint method [Arridge, 1999] with Rytov normalisation:

$$J_{c,n} = \frac{-\Phi_s(n)\,\Phi_d(n)\,V_n}{|\mathbf{R}_d^\top \boldsymbol{\Phi}_s|} \tag{5}$$

where c indexes the source–detector channel, n the mesh node, Φ_s and Φ_d are the forward and adjoint Green's functions, and V_n is the nodal volume. Channels with source–detector distance outside 10 mm to 55 mm are excluded.

Regularisation. The regularisation parameter λ is selected by minimising the Generalised Cross-Validation (GCV) criterion:

$$\mathrm{GCV}(\lambda) = \frac{\|\mathbf{J}\hat{\mathbf{x}} - \mathbf{d}\|^2/m}{\left(\mathrm{tr}(\mathbf{I} - \mathbf{H}_\lambda)/m\right)^2} \tag{6}$$

where $\mathbf{H}_\lambda = \mathbf{J}(\mathbf{J}^\top\mathbf{J} + \lambda\mathbf{I})^{-1}\mathbf{J}^\top$ and m is the number of measurements. An L-curve corner detection method is also available.

Tikhonov inversion. Per-timepoint reconstruction: $\Delta\hat{\mu}_a = (\mathbf{J}^\top\mathbf{J} + \lambda\mathbf{I})^{-1}\mathbf{J}^\top\Delta\mathrm{OD}$.

3.6 Spectral unmixing

Absorption changes at two wavelengths are decomposed into chromophore concentration changes via the pseudoinverse of the extinction matrix:

$$\begin{pmatrix} \Delta[\mathrm{HbO_2}] \\ \Delta[\mathrm{Hb}] \end{pmatrix} = \mathbf{E}^+ \begin{pmatrix} \Delta\mu_a(\lambda_1) \\ \Delta\mu_a(\lambda_2) \end{pmatrix} \tag{7}$$

where $\mathbf{E}$ contains molar extinction coefficients from the Prahl/OMLC tables [Prahl, 1999] interpolated to 750 and 850 nm.

3.7 Spatial smoothing and normalisation

Reconstructed images are spatially smoothed with a Gaussian kernel (FWHM = 13 mm) applied on the mesh nodes, matching the published NIfTI kernel. Time series are z-scored per node for scale-invariant comparison.

4 Results

4.1 Cross-validation against redbirdpy

Table 2: dot-jax test suite: 272 tests across 11 modules.

Module	Tests	Validation scope
`analytical`	32	Bessel, harmonics, fluence, reflection
`property`	34	Extinction, Beer-Lambert, scattering
`mesh`	37	Volumes, areas, deldotdel, surface extraction
`assembly`	33	K, M, C matrices vs. redbirdpy
`forward`	25	Source location, adjoint, dense/sparse
`spectral`	10	Multi-wavelength, Jacobian
`recon`	13	L-curve, GCV, Gauss-Newton
`hemodynamics`	45	OD, motion, pruning, regression, GVTD
`io`	25	SNIRF, BIDS-fNIRS, ds004569
`atlas`	12	MNI152 mesh, optode projection
`types`	6	Constants, containers
Total	**272**	

All assembly, analytical, and property functions are cross-validated against redbirdpy to relative tolerance 10^{-10}, confirming numerical equivalence of the JAX reimplementation. The system matrix $\mathbf{A}$ matches redbirdpy's `femlhs` output to machine precision on the test mesh.

4.2 HD-DOT processing: sub-02, session 01

Table 3: HD-DOT processing pipeline results for sub-02/ses-01.

Preprocessing	
Raw channels	3684 (750/850 nm)
Channels after pruning	3563 (121 rejected)
Motion artifacts	0.02% of samples
Forward model	
Atlas mesh	3000 nodes, 14820 elements
Jacobian (750 nm)	(1158, 3000)
Jacobian (850 nm)	(1139, 3000)
λ_{GCV} (750 nm)	2620
λ_{GCV} (850 nm)	442
Reconstruction	
$\Delta\mu_a$ range (750 nm)	[-0.019, 0.022] /mm
$\Delta\mu_a$ range (850 nm)	[-0.043, 0.042] /mm
ΔHbO_2 range (smoothed)	[-98, 87] μM

4.3 Comparison with published NIfTI reconstruction

We compare z-scored dot-jax HbO_2 reconstructions to the published NeuroDOT NIfTI images at mesh nodes that fall within the NIfTI brain mask (1267 active nodes of 3,000).

Table 4: Correlation between dot-jax and NeuroDOT HbO_2 reconstructions (z-scored).

Metric	Value	Notes
Global mean temporal r	0.031	Averaged over 1,267 nodes
Mean spatial r (per timepoint)	0.003	641 timepoints
Median spatial r	0.003	
Mean temporal r (per node)	0.003	1,267 nodes
Best node temporal r	0.172	
Nodes with $r > 0.1$	17/1,267	1.3%

The weak but positive correlations are expected given the fundamental differences between the two pipelines:

1. **Generic vs. subject-specific forward model.** dot-jax uses a standard MNI152 atlas mesh with population-average optical properties, while NeuroDOT uses a calibrated subject-specific "Amat" sensitivity matrix.

2. **Mesh resolution.** Our 3,000-node atlas mesh is far coarser than the NeuroDOT volumetric grid ($48 \times 64 \times 48$ voxels).

3. **Reconstruction algorithm.** Tikhonov-regularised pseudoinverse vs. NeuroDOT's optimised reconstruction with spatiotemporal priors.

4. **Coordinate system.** MNI152 vs. Talairach, with potential residual misalignment at the mesh-to-voxel mapping.

5 Discussion

dot-jax demonstrates that the complete CW-DOT forward–inverse pipeline can be implemented as a differentiable, JIT-compiled JAX program. The key contribution is not the reconstruction accuracy — which, on a generic atlas with default parameters, cannot match calibrated subject-specific systems — but the **architectural capability**: every function in the pipeline, from optical property computation through system matrix assembly to detector value extraction, is composable with `jax.grad`, `jax.vmap`, and `jax.jit`.

This opens several directions:

Automatic Jacobian verification. The adjoint Jacobian formula $J = -\Phi_s \otimes \Phi_d \odot V$ can be independently verified by computing $\partial \hat{y} / \partial \mu_a$ via `jax.grad` through the full solve. dot-jax is, to our knowledge, the first DOT package where this cross-check is available.

Gradient-based parameter optimisation. Optical properties (μ_a, μ_s', n), regularisation parameters, and even mesh node positions can be optimised by gradient descent through the forward model. This is the foundation for learned forward models and neural-ODE-based reconstruction.

Simulation-based inference for DOT. The JIT-compiled forward model can serve as the simulator for SBI [Cranmer et al., 2020], enabling amortised posterior estimation over optical property maps. A single trained neural density estimator could replace per-subject iterative reconstruction.

Integration with the NeuroJAX ecosystem. dot-jax shares the JAX/Equinox/Lineax stack with vpjax (hemodynamic modelling), sbi4dwi (diffusion microstructure), and neurojax (whole-brain dynamics). This enables joint forward models where diffusion-derived tissue structure constrains optical scattering, hemodynamic models couple to the DOT observation equation, and neural dynamics drive the chromophore concentration changes that DOT observes.

5.1 Limitations

The current implementation uses dense matrix assembly, limiting practical mesh sizes to ∼10,000 nodes on CPU. The sparse CG solver (`forward_cw_sparse`) extends this to ∼100,000 nodes on GPU. Subject-specific head meshes from segmented MRI would substantially improve reconstruction accuracy. The absence of short-separation channels in the ds004569 cap design limits superficial signal regression.

6 Software

dot-jax is open source under GPL-3.0: `https://github.com/m9h/dot-jax`

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Table 5: Software dependencies and versions.

Package	Version
JAX / jaxlib	$\geq$ 0.4.30
Equinox	$\geq$ 0.11.0
Lineax	$\geq$ 0.0.5
NumPy	$\geq$ 1.26
SciPy	$\geq$ 1.10
h5py	$\geq$ 3.8
redbirdpy	0.2.1 (cross-validation)
Python	3.11–3.13

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